Usefulness of translational equivariance

- This idea works **extremely well**. **ImageNet** is an image classification task with 14 million images, 20K categories.

ImageNet Classification with Deep Convolutional Neural Networks

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Imagenet classification with deep convolutional neural networks
A Krizhevsky, I Sutskever… - Advances in neural …, 2012 - proceedings.neurips.cc
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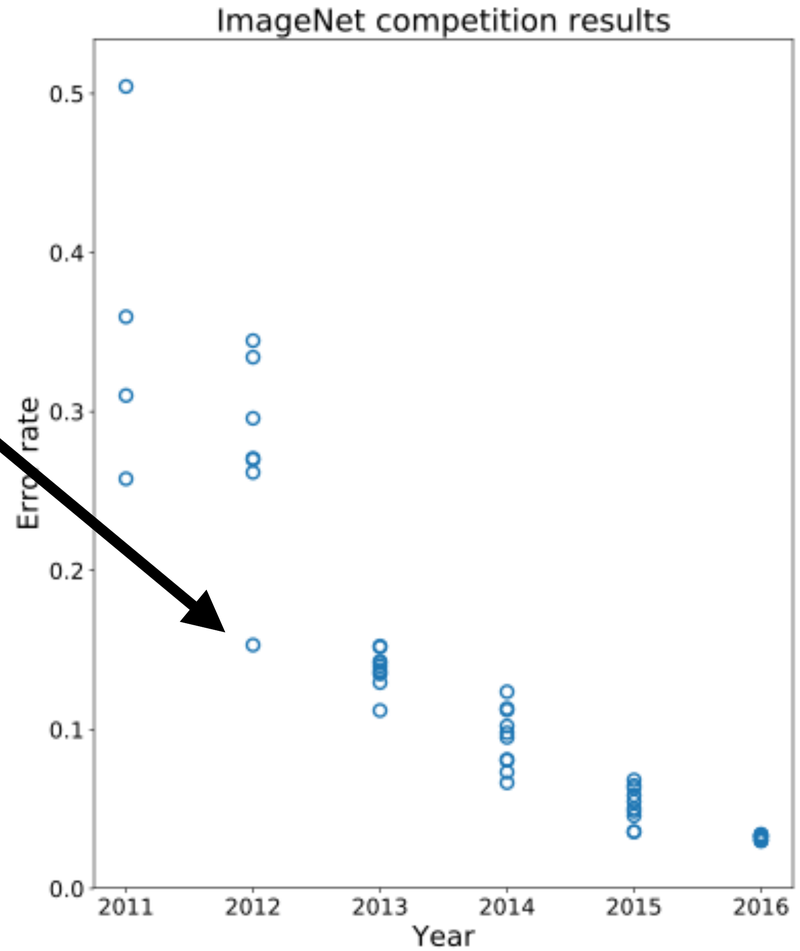
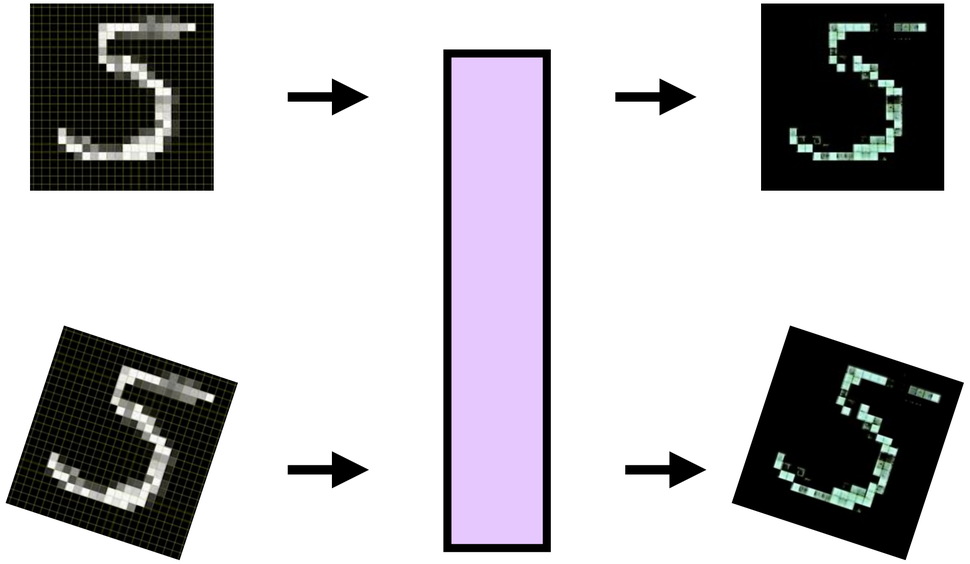


- This result helped start the deep learning revolution.

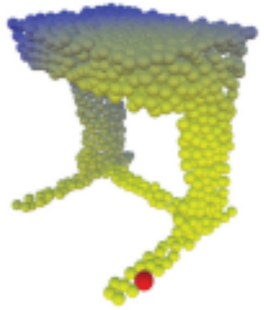
Geometric deep learning

- The key idea here is that constraining the network with translational symmetry dramatically narrows down the space of functions we optimize over, focusing on useful ones. **“Inductive bias”**.

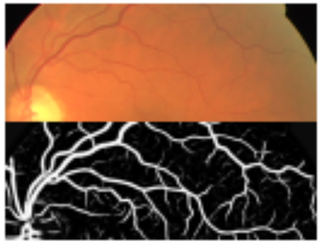
- CNN was for translations. It turns out we can do this for **any symmetry group *G*** (Cohen + Welling 2016): **group-equivariant neural networks.**



This framework — **geometric deep learning** — is a very useful thing to do! (Extensive formalism exists).



Wang et. al, 2019



Bekkers et. al, 2018

Hoogeboom et. al, 2022